MARCH, 2026

REACH FOR CHANGE

PREDICTING DONOR RESPONSE TO OPTIMIZE OUTREACH FOR SOCIAL GOOD

DATA MINING II
GROUP PROJECT 2025/2026

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I. INTRODUCTION

In recent years, nonprofit organizations have faced an unexpected challenge.

While the number of charitable causes has grown significantly, people's tolerance for repeated solicitations has decreased. Many potential supporters report feeling overwhelmed by generic fundraising campaigns, leading to lower engagement and long-term disengagement from charitable initiatives.

To address this issue, the ***Civic Support Alliance (CSA)***, a federation representing multiple humanitarian and social aid programs, has decided to modernize how outreach campaigns are conducted.

Instead of contacting everyone in their database, the organization wants to contact individuals who are most likely to respond positively. The objective is not only operational efficiency but also respect, contacting fewer people but contacting the right people.

The *CSA* has accumulated historical data from previous fundraising campaigns, including demographic information, past interaction history, and donation behavior. They have provided a dataset containing individuals who may be contacted in an upcoming campaign, along with the outcome of a past campaign.

Your team has been hired as a group of data scientists to help **design the predictive system** that will guide the next outreach effort. Your goal is to build a predictive model capable of answering the following question: ***Will this person donate if contacted?***

II. PROJECT GOALS

The primary components and goals of the project are:

1. **Preprocessing and Exploratory Data Analysis (EDA):** Clean and preprocess your dataset, performing an exploratory analysis to uncover patterns and relationships.
2. **Binary Classification & Optimization:** Create a classification model to accurately predict if a certain person will donate. To do that, you will need to **develop a consistent model assessment** strategy to compare different candidate models and find the most generalizable one. During your selection of best (or set of best) model(s), you are encouraged to **explore ways to improve their performance** (e.g., hyperparameter tuning or preprocessing/feature selection adjustments). Don't forget to report all models, compare the optimized model with the previous models and discuss the findings.
3. **Kaggle Competition:** Teams can submit multiple predictions on Kaggle, with the scoreboard ranking submissions based on **binary F1 Score**. Before the competition ends, you must choose one submission to compete (check Kaggle tutorial on Moodle).
4. **Additional Insights:** This component is **open-ended**, allowing you to explore a wide range of ideas of your preference, as long as it relates with the topic/dataset, and you can clearly communicate them. Here are some suggestions:
    a. Analyze and discuss the importance of the features how they contribute towards the prediction.
    b. Analyze and discuss the incorrect predictions.
    c. Analyze the probability of your predictions and the calibration of the model.
    d. Create an analytics interface that returns a prediction when new input data is provided.

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III. DATASET

You have access to two different datasets:

In the **training set (donors_*train.csv*)**, you will find features that give detailed information about each possible donor. Use this training data and its features to build and validate your machine-learning models. **Important note: You should not consider the target variable as feature for any of the predictive models.**

In the **test dataset (donors_test.csv)**, you can still access the same descriptive attributes from an independent set of possible donors. You must use the models trained with the training set to make a prediction for all observations in this dataset. **You will not have access to the target value on this set**, but you will be able to know how well your model performs on this data by submitting your prediction on Kaggle. You'll find the example of a (random) submission in *sample_submission.csv*.

The available data contains the following attributes:

ATTRIBUTE	DESCRIPTION
TARGET_B	Whether person has donated in the last year's campaign or not (1 or 0 respectively)
CONTROL_NUMBER	Unique identifier of each possible donor
DONOR_AGE	Age as of last year's mail solicitation
URBANICITY	Nature of the donor's location U = Urban C = City T = Town R = Rural S=Suburban
SES	Code assigning each individual to one of 5 demographic profiles based on socioeconomic characteristics
HOME_OWNER	H if individual is a Homeowner, U if it is Unknown
DONOR_GENDER	Gender of individual
INCOME_GROUP	Code assigning each individual to one of 7 income group levels
WEALTH_RATING	Code assigning each individual to one of 10 wealth rating groups
MEDIAN_HOME_VALUE	Individuals' median home value in $100's
MEDIAN_HOUSEHOLD_INCOME	Individuals' median household income in $100's
PCT_OWNER_OCCUPIED	Percent of owner-occupied housing near where person lives
PER_CAPITA_INCOME	Per Capita Income of neighborhood in which person lives
PCT_ATTRIBUTE1	Percent of residents in person's neighborhood that are male and active military
PCT_ATTRIBUTE2	Percent of residents in person's neighborhood that are male and veterans
PCT_ATTRIBUTE3	Percent of residents in person's neighborhood that are Vietnam veterans

ATTRIBUTE	DESCRIPTION
PCT_ATTRIBUTE4	Percent of residents in person's neighborhood that are WW2 veterans
PEP_STAR	If person achieved STAR donor status, given to individuals who have given to 3 consecutive campaigns.
RECENT_STAR_STATUS	If achieved STAR within last 4 years.
RECENCY_STATUS_96NK	Donor status: **A** (**active** - made their first donation more than 12 months ago and has made a donation in the last 12 months.); **E** (**inactive** - made a donation 25+ months ago.); **F** (**first time** - made their first donation in the last 6 months and has made just one donation.); **L** (**lapsing** - made their last donation between 13-24 months ago); **N** (**new** - made their first donation in the last 12 months and is not a First time donor); **S** (**star** donor)
FREQUENCY_STATUS_97NK	Number of times donated in last period (The period of recency for all groups except L and E is the last 12 months. For L it is 13-24 months ago, and for E it is 25-36 months ago).
RECENT_RESPONSE_PROP	Proportion of responses to the individual to the number of (card or other) solicitations from the charitable organization in the last four years
RECENT_AVG_GIFT_AMT	Average donation from the individual to the charitable organization in the last four years.
RECENT_CARD_RESPONSE_PROP	Number of times the individual has responded to a card solicitation from the charitable organization in the last four years
RECENT_AVG_CARD_GIFT_AMT	Average donation from the individual in response to a card solicitation from the charitable organization in the last four years
RECENT_RESPONSE_COUNT	Number of times the individual has responded to a promotion (card or other) from the charitable organization in the last four years
RECENT_CARD_RESPONSE_COUNT	Number of times the individual has responded to a card solicitation from the charitable organization in the last four years.
MONTHS_SINCE_LAST_PROM_RESP	Number of months since the individual has responded to a promotion by the charitable organization
LIFETIME_CARD_PROM	Total number of card promotions sent to the individual by the charitable organization
LIFETIME_PROM	Total number of promotions (card or other) sent to the individual by the charitable organization
LIFETIME_GIFT_AMOUNT	Total lifetime donation amount from the individual to the charitable organization
LIFETIME_GIFT_COUNT	Total number of donations from the individual to the charitable organization
LIFETIME_MAX_GIFT_AMT	Maximum donation amount from individual
LIFETIME_MIN_GIFT_AMT	Minimum donation amount from individual
LAST_GIFT_AMT	Amount donated in the individuals most recent donation

ATTRIBUTE	DESCRIPTION
CARD_PROM_12	Number of card promotions sent to the individual by the charitable organization in the last 12 months
NUMBER_PROM_12	Number of promotions (card or other) sent to the individual by the charitable organization in the last 12 months
MONTHS_SINCE_LAST_GIFT	Number of months since the most recent donation from the individual to the charitable organization
MONTHS_SINCE_FIRST_GIFT	Number of months since the first donation from the individual to the charitable organization
FILE_CARD_GIFT	Lifetime average donation from the individual in response to all card solicitations from the charitable organization
CHILDREN	Number of children the individual has

IV. OUTLINE

Your project deliverables (notebook or zip of notebooks) should respect the outline:

Group Member Contribution

What part(s) of the work were done by each member and an estimated % contribution of each member towards the final work.

Abstract

Provide a small overview of your work (200 to 300 words): What is the context? What are your goals? What did you do? What were your main results, and what conclusions did you draw from them?

I. Introduction

- Overview and main goals of the project.
- Description of the overall process and identification of model assessment approach adopted in the work (CV, Holdout, etc...)

II. Data Exploration and Preprocessing

- Description of data received -> key insights
- Steps taken to clean and prepare the data based on exploration

III. Binary Classification & Optimization

- Explanation of model assessment strategy and metrics used
- Feature selection strategy and results
- Optimization efforts, results and discussion
- Comparison of performance between candidate models

IV. Deployment

The final section of your work should deploy your modeling strategy to generate reliable predictions for new data. The output should be the .csv file that you consider the solution you selected on Kaggle as your best.

V. Open-Ended Section

- Objectives for the section and description of the actions taken
- Results and discussion of main findings -> key takeaways

Note: This section expects that the objectives set go beyond what would reasonably be considered as adding or removing techniques to your pipeline.

V. DELIVERABLES

Upon the project's deadline, you will be required to submit:

- A *Jupyter* notebook (or a zipfile of notebooks) featuring all the code you used throughout the project to:
  a) Decide on your final solution
  b) Obtain your final results (code that helped you make decisions, but does not directly contribute to reaching the goal, should be included, but commented).

- Make sure to include detailed markdown cells and comments within the notebook to guide readers through the code. Clearly explain the purpose of each code segment, the insights gained, and the decisions made.
- Name your file in the format ***DM2_GroupXX***, where ***GroupXX*** should be your group number. This naming format should also be used for your Kaggle submissions.

VI. EVALUATION

Your work will be evaluated according to the following criteria:

CRITERIA	PERCENTAGE (%)	MAXIMUM GRADE (OUT OF 20)
Project Structure & Notebook (s) Quality	20	4
Preprocessing and EDA	20	4
Modeling and Optimization	30	6
Model performance on Kaggle	10	2
Open-Ended Section	10	2
Discussion Organization & Performance	10	2

- Groups can submit all deliverables with a maximum delay of **3 days**, incurring a penalty of **1 point per day**. Beyond these three days, submissions will not be accepted.
- Deliveries made before the deadline will receive a bonus of **0.15 points per day** of delivery in advance (up to a **maximum of 1 point**).

Deadline: June 5th, 2026, at 18:00

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Your grade will reflect our assessment of the quality of your work in different aspects. Please find below more details about what is taken into consideration for each topic:

- **Project Structure and Notebook(s) Quality (4v):** Your notebook(s) should be readable and understandable to someone reading and looking at them for the first time. Every cell you present should have a clear purpose. This section also encompasses the overall quality of your introduction and conclusions for each cell.
- **Preprocessing and EDA (4v):** Describe the dataset and extract meaningful insights that you consider relevant to the problem. Avoid adding unnecessary visualizations or elements. Additionally, it should clearly explain the steps and rationale behind the data cleaning process to make the data usable by your predictive models.
- **Modeling and Optimization (6v):** Describe your strategy for the classification objective. This section is separated into different components:
  - Additional preprocessing
  - Feature selection strategy and Implementation
  - Modelling approach – assessment strategy (holdout, cross-validation, etc.) and the algorithms explored (minimum of 4 covered in class)
  - Performance assessment – rationale for choice of evaluation metric(s) and interpretation of results
  - Model optimization
- **Model performance on Kaggle (2v):** Grading based on groups ranking.
- **Open-Ended Section (2v):** Describe your strategy for the additional insight's objective. This section is separated into different components:
  - Formulation and adequacy of the Objectives
  - Difficulty of tasks
  - Correctness/efficiency of implementation
  - Explanation and discussion of results
  - Alignment between results and communicated objectives
- **Discussion Organization & Performance (2v):** The delivered notebooks should be developed with the final presentation in mind.

During the final discussion, each group member will be randomly selected to present and explain a portion of the notebook. Therefore, the notebook must be clear, well-structured, and thoughtfully organized to ensure a coherent and professional presentation to the evaluators.
It is the group's responsibility to ensure that the notebook is fully presentation-ready. This includes clearly stating the goal, approach, and main takeaways at the beginning of each section; providing readable, well-commented code; and clearly highlighting the key insights and results throughout.

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VII. PARTING NOTES

- If you use automated packages (e.g., ydata, pandas-profiling), we will only consider observations that have a visible impact on the project (i.e., a key insight that is acted upon).
- Using Lazy Predict or similar AutoML (e.g., Feature Tools, TSFresh) packages, XGBoost, CatBoost, LightGBM (or similar packages not in vanilla sk-learn) is explicitly off-limits and will result in a 1-point penalty per attempt.
- We won't consider any steps or results that are not mentioned in any of your submitted notebooks.
- Everything in your clean notebook must have a clear purpose. Avoid including irrelevant, unimportant, or redundant information. Additionally, please refrain from providing theoretical explanations of topics covered in class.
- The trustworthiness of the information you provide is key. You should look to source information from peer-reviewed journals (thus, avoid citing Medium, TowardsDataScience and similar sources). Also, for the love of God, DO NOT CITE ChatGPT or any other LLM for that matter.
- Before submitting, run your notebook from the start one last time (if you used a GridSearch, you can comment this cell, but you should run the final model with the GS parameters in a different cell).
- We will run your Jupyter Notebooks. Please ensure that we can run the notebook from start to finish without interruption. Notebooks that do not fulfil this condition will be severely penalized.
- The code will pass through a process of plagiarism and AI generation checking.
- You must submit your predictions on the Kaggle competition before the project deadline to get points for that component. Submissions must be identified with same format as your deliverable (see Section V). Submissions not clearly identified can't be considered/designated to the corresponding group.
- When determining the grade for your work, there will be a comparative component between it and the work presented by your peers.

Friendly Reminders:

- Attendance at the **defense is mandatory for approval** in the project. The order in which the group members present their notebooks is decided by the professors at the start of the discussion. Not attending will result in a 0 in the project.
- A student's final grade can change during the defense depending on their performance**, without any limitations.**
- If something is good enough to be mentioned, it is also good enough to know. DO NOT include techniques/algorithms/steps you cannot explain in your work: we may (and probably will) ask about them in the discussion.
- Finished is better than perfect.